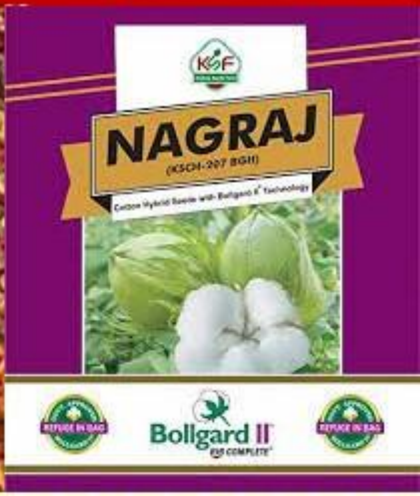
INDIAN ECONOMY

LEVEL II



LECTURES

- A BRIEF ON HYDROPONICS
- A BRIEF ON PPV & FR ACT, 2001
- A GLANCE AT THE SEEDS ACT, 1966

MODULE - 72

WHAT DO WE NEED FOR PLANTS TO GROW?

- Sunlight
- Oxygen
- Water
- Nutrients

$\left.\begin{array}{l}\text{Water}\\ \text{Nutrients}\end{array}\right\}$ Soil

WHY DO WE NEED SOIL?

- Soil provides hospitable place for a plant to anchor its roots and serves as a source of water and nutrients needed for its growth.
- If these two purposes are met by other means, plants would not need soil. Then, we arrive at the basic principle behind hydroponic farming.

HENCE, WHAT IS HYDROPONICS?

- It is the technique of growing plants without soil.
- In traditional farming, plants get their nutrients from soil, through additives such as compost, manure and fertilizers.
- In hydroponic farming, plants get them in nutrient fortified water.

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WHAT IS THE MOST COMMON METHOD OF HYDROPONIC FARMING?

The method is catching up in urban areas in India, where households meet some of their vegetable needs this way.

- The most common method is to place the plants in a plastic trough, their roots are dipped directly into nutrient rich solutions.
- The roots can be supported by a medium such as rockwool or peat moss, which acts as a sterile substitute for soil.
- These are watered with nutrient solution, it can be done indoors but proper monitoring of pH level of water, temperature and amount of light the plants receive is required.
- The container is to be oxygenated using tools such as an air pump.

HYDROPONIC FARMING … ADVANTAGES

FASTER GROWTH IN CONTROLLED ENVIRONMENT!

1 There is no need to worry about land requirement.

2 Roots don’t need to spread because water and nutrients are delivered directly to them, hence more plants can be grown in the same amount of space.

3 More energy can be diverted into the growth of leaves, stems, vegetables and fruits and hence, 30% to 50% faster growth can be achieved.

4 We can produce food anywhere in the world at anytime of the year as climate and light can be controlled. Food grown in this way is nutritionally superior.

5 The risks of pests will reduce substantially.

HYDROPONIC FARMING ... ADVANTAGES

7 Global human population is expected to reach around 10 billion somewhere in between 2050 and 2080, whereas in 2019 alone about 124 million people faced acute food shortages from climate related events. Hence, it is seen as a sustainable solution towards food security.

FASTER GROWTH IN CONTROLLED ENVIRONMENT!

6 Hydroponics use less water than traditional soil based systems. Moreover, the water can be filtered, repopulated with nutrients and fed back to plants. The same water can be used over and over again. Overall, it reduces water dependence by as much as 90%.

HYDROPONIC FARMING … BUT, WHAT ABOUT THE CHALLENGES?

EXPENSIVE AND NEED FOR CONTINUOUS MONITORING!

1. Setting up an hydroponic farm can be extremely expensive due to the cost of containers, pumps, lights, nutrients and automated systems.

2. Since plants are grown in a controlled environment, constant monitoring is required.

3. The process of hydroponic farming depends on a range of equipment that requires proper expertise.

4. High level of energy consumption, because of air pumps, providing light, water pumps and the running of other appliances.

5. Normally, we cannot grow root vegetables through hydroponics.

6. Heavy fruiting plants may need elaborate forms of support.

Overall, this appears to be the future, when one looks at sustainable solution for the food shortages.

Learning Space

FARMERS' RIGHTS UNDER PPV&FR ACT, 2001 AND INDIAN PATENTS ACT, 1970

The PPV&FR Act is the first of its kind in granting intellectual property rights not only to plant breeders but also to the farmers by protecting new, extant and farmers' varieties.

- As per Section 39 (1) (iv) of The Protection of Plant Varieties and Farmers' Rights Act, 2001, a farmer is allowed to save, use, sow, resow, exchange, share or sell his farm produce, including seed of a variety protected under this Act. He is not punishable as long as, he does not sell "Branded Seed".
- Hence, farmers have rights to produce seeds and even sell seeds of a protected variety, provided it is unbranded.
- As per the Indian Patents Act, a biological process to create a seed or plant cannot be patented.
- Hybridization, which is the process of interbreeding between individuals of different species, comes under biological process and cannot be patented.

PPV & FRA, 2001 MAY NOT RESULT IN VIBRANT AGRICULTURAL ECOSYSTEM!

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MANY THINGS ARE UNCLEAR!

1 The question of what exactly the rights granted under the PPV & FRA is not clear. Is it the equivalent of a patent or is it a right to produce a variety or is it a right over production from a variety?

2 The private sector now accounts for around 65 per cent of seed production in India and is an essential part of the agri-innovation chain. Without proper IP protection, probably, it is very difficult for Indian agriculture to move forward with innovation by development of high yielding varieties.

3 Government research laboratories and Universities, normally concentrate on key food crops. Private sector role is significant, when one looks at horticulture crops like fruits/vegetables, plantation crops, oil seeds, commercial crops etc.

4 There is widespread variance in state level laws and regulations and this makes the things quite complex.

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WHY IS INDIA NOT A MEMBER OF UPOV?

WHAT IS UPOV?

The International Union for the Protection of New Varieties of Plants (UPOV) is an intergovernmental organization with headquarters in Geneva, Switzerland. UPOV's mission is to provide and promote an effective system of plant variety protection.

- In India, farmers' interests far outweigh the breeders interests. UPOV primarily looks at the rights of plant breeders.
- Though UPOV offers limited rights to the farmers, the countries subscribing to UPOV have to uphold commercial interests of the seed developers.
- Hence, India did not join UPOV as a member, as farmers' interests are paramount in India.
- However, there is constant pressure on India from advanced countries to subscribe to UPOV.

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WHAT DO YOU KNOW ABOUT SEEDS ACT, 1966?

Seeds Act, 1966 has labelling provisions, but no licensing provisions.

- Almost at the time of 'Green Revolution', India brought Seeds Act, 1966.
- It was the first Act to govern matters of seed and seed quality. It covers "notified kinds or varieties of seeds".
- Regulation of quality is limited to the seeds of varieties that have been officially notified. Such varieties are mostly those bred by public sector institutions like, ICAR State Agricultural Universities (SAUs) etc.
- These are officially released for cultivation after multi-location trials, over three years or more, to evaluate their yield performance, disease and pest resistance.

ARE THE PRIVATELY-BRED HYBRIDS COVERED UNDER ANY REGULATION?

Kaveri Seeds

NUZIVEEDU SEEDS

1. As already stated, the current Seeds Act, 1966 is applicable only to the notified varieties.

2. Unless a variety or hybrid is notified, its seeds cannot be certified.

3. Most of the private hybrids marketed in India, by virtue of not being officially "released", are neither "notified" nor "certified". Instead, they are "truthfully labelled", the companies sell them with the label.

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WHAT DO YOU UNDERSTAND BY 'TRUTHFUL LABELLING'?

WHAT ABOUT THE RISKS?

At present, there are risks to the farmers getting low-quality seeds, especially by fly-by-night operators with "truthful labelling" and "self-certification" processes.

1 The companies selling the seeds simply state that, seeds inside the packets have a minimum germination. Say for example 80% means - if 100 are sown, at least 80 will produce plants.

2

- Truthful Labelling talk about ...
  - ✓ Genetic purity, that means, percentage of "true-to-type" plants and non-contamination by genetic material of other varieties/species.
  - ✓ Physical purity, proportion of non-contamination by other crop/weed seeds or inert matter.

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WHAT IS INTENDED IN THE NEW SEEDS BILL, 2019?

1. It provides for compulsory registration of any kind or variety of seeds that are sought to be sold.

2. No seed of any kind or variety, for the purpose of sowing or planting shall be sold, unless such kind or variety is registered. Even hybrids/varieties of private companies are to be registered.

3. They have to meet the minimum prescribed standards relating to germination, physical and genetic purity.

4. Breeders would be required to disclose the expected performance.

5. If the seed of such registered kind or variety fails to provide the expected performance under such given conditions, the farmers can approach under the Consumer Protection Act.

Till the end of 2022, the bill could not be passed! There was huge opposition from the farmers' groups.

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SINCERE THANKS
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